FRONTEND CURRICULAM

HTML – BASICS TO ADVANCED

1. HTML5- Basics to Advanced
    a. HTML- Introduction
    b. HTML- Code Editor Setup
    c. HTML- Structure of an HTML document
    d. HTML - Adding Heading in a Webpage
    e. HTML - Adding Paragraph in a Webpage
    f. HTML - Adding Images in Web Pages
    g. HTML - Creating Ordered and Unordered Lists
    h. HTML - Adding Styles and Formatting HTML Elements
    i. HTML - Block and Inline Display Values
    j. HTML - Adding group styles using Class attribute
    k. HTML - Adding HyperLinks in Webpages
    l. HTML - Using links to connect Webpages
    m. HTML - Uniquely Identify HTML Element by using ID attribute
    n. HTML - What makes HTML5 Better?
    o. HTML - New Elements & Attributes in HTML5
    p. HTML - Design Page Structure Using Semantic Elements
    q. HTML - New Semantic Elements: <details> and <summary>
    r. HTML - Introduction to Tables
    s. HTML - Span table rows and columns
    t. HTML - Design layouts using Tables
    u. HTML Tables - Practice Problem 1
    v. HTML Tables - Practice Problem 1: Solution
    w. HTML - Introduction to Forms
    x. HTML - Form Input Types
    y. HTML - Adding Styles to Forms
    z. HTML - Form Methods: GET and POST
    aa. HTML - New Form Elements
    bb. HTML - New Form Input Types
    cc. HTML - New Form Input Attributes
    dd. HTML - Form Validation
    ee. HTML - Forms Practice Problem 1
    ff. HTML - Forms Practice Problem 1 - Solution
    gg. HTML - Adding Videos in a Webpage
    hh. HTML - Adding Audio in a Webpage
    ii. HTML - The <embed> element

jj. HTML - Embed Youtube Videos in a Webpage
kk. HTML - Introduction to SVG
ll. HTML - Create circle using SVG
mm. HTML - Create a rectangle using SVG"

Problem 1: Favorite Foods Webpage

Problem 2: Table Practice

Problem 3: Forms Practice

CSS- BASICS TO ADVANCED

1. CSS2- Basics to Advanced
    a. CSS - Introduction
    b. CSS - How to use CSS?
    c. CSS - Styling Text
    d. CSS - Adding Borders
    e. CSS - Margin and Padding
    f. CSS - Styling Lists
    g. CSS - Styling Links
    h. CSS - Styling Buttons
    i. CSS - Adding Icons
    j. CSS - Working with Display Values
    k. CSS - Display value: Inline-Block
    l. CSS - Positioning HTML Elements
    m. CSS - Handling Content Overflow
    n. CSS - Floating HTML Elements
    o. CSS - Box Model
    p. CSS - Box Sizing
    q. CSS - Adding Shadows
    r. CSS - Combinators
    s. CSS - Pseudo Classes
    t. CSS - Pseudo Elements
    u. CSS - Adding Backgrounds
    v. CSS - Gradients
    w. CSS - Handling Text Overflow
    x. CSS - Introduction to Responsive Web Design
    y. CSS - Viewport
    z. CSS - Dynamic Size
    aa. CSS - Introduction to Media Queries
    bb. CSS - Responsive Webpage Design using Media Queries
    cc. CSS - Flex Layout
    dd. CSS - Flex Container Properties

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ee. CSS - Align Flex Items
ff. CSS - Flex Item Properties
gg. CSS - Design Topbar using Flex
hh. CSS - Responsive Grid using Flex
ii. CSS - Transform Property
jj. CSS - Align Elements using Transform Property
kk. CSS - Transition Property
ll. CSS - Animation Property
mm. CSS - Flying Rocket Animation using Animation Property
nn. CSS - Introduction to Grid Layout
oo. CSS - Grid Container Properties
pp. CSS - Grid Item Properties

Problem 1: Create Google.com Look-Alike

Problem 2: Final Project Problem Statement

JAVASCRIPT

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2. Javascript
    a. JavaScript - Introduction
    b. JavaScript - Introduction to Variables, Operators and Expressions
    c. JavaScript - Variables
    d. JavaScript - Numbers
    e. JavaScript - In-Built Number Functions
    f. JavaScript - Strings
    g. JavaScript - In-Built String Functions
    h. JavaScript - Null and Undefined Values
    i. JavaScript - Conditional Statement: If-else
    j. JavaScript - Conditional Statement: Switch
    k. JavaScript - Arithmetic Operators
    l. JavaScript - Assignment Operators
    m. JavaScript - Comparison and Logical Operators
    n. JavaScript - Implicit Type Coercion
    o. JavaScript - Explicit Type Coercion
    p. JavaScript - Objects
    q. JavaScript - Arrays Part
    r. JavaScript - Introduction to Loops
    s. JavaScript - While Loop
    t. JavaScript - For Loops
    u. JavaScript - Functions
    v. JavaScript - Function Arguments
    w. JavaScript - Scope and Environment

x. JavaScript - Variable Hoisting
y. JavaScript - Function Hoisting
z. JavaScript - Introduction to DOM
aa. JavaScript - Select HTML Elements
bb. JavaScript - Query Selectors
cc. JavaScript - Update HTML Elements
dd. JavaScript - Updating Styles
ee. JavaScript - Intro to Events
ff. JavaScript - Add and Remove Classes from HTML Elements
gg. JavaScript - Getting Styles
hh. JavaScript - Form Events
ii. JavaScript - Keyboard Events
jj. JavaScript - Mouse Events
kk. JavaScript - Create HTML Elements
ll. JavaScript- Update and Delete HTML Elements
mm. JavaScript - Introduction to Browser Object Model
nn. JavaScript - Screen and Navigator Object
oo. JavaScript - History and Location Object
pp. JavaScript - Mini-Project 1: Smartwatch Showcase
qq. JavaScript - Mini-Project 2: Calculator App
rr. JavaScript - Introduction to Regular Expressions
ss. JavaScript - Meta-characters and Character Classes
tt. JavaScript - Regex Quantifiers
uu. JavaScript - Using Regex inside JavaScript
vv. JavaScript - Regex Groups
ww. JavaScript - Form Validation using Regex
xx. JavaScript - Introduction to Object Oriented Programming
yy. JavaScript - The ""this"" Keyword
zz. JavaScript - Constructor Functions
aaa. JavaScript - Inheritance and Prototype Chain
bbb. JavaScript - Prototype Implementation
ccc. JavaScript - Understanding Frontend and Backend
ddd. JavaScript - HTTP Request and Response
eee. JavaScript - Introduction to JSON
fff. JavaScript - Introduction to AJAX
ggg. JavaScript - Handling HTTP Response
hhh. JavaScript - Working with HTTP Response Data
iii. JavaScript - Making HTTP Post Requests"

Problem 1: Smartwatch Showcase

Problem 2: Calculator App

JQUERY

1. jQuery
    a. jQuery - Introduction
    b. jQuery - Selectors and Filters
    c. jQuery - Add/Update HTML Elements
    d. jQuery - Add/Update Styles
    e. jQuery - Handling HTML Events
    f. jQuery - Making AJAX Calls

REACT

1. Modern Javascript (ES6) for React
    a. ES6 - Introduction
    b. ES6 - Variable Creation using ""let"" and ""const""
    c. ES6 - Template Strings
    d. ES6 - Arrow Functions
    e. ES6 - Rest Operator
    f. ES6 - Spread Operator
    g. ES6 - Destructuring
    h. ES6 - Array Functions: map()
    i. ES6 - Arrow Functions: reduce()
    j. ES6 - Arrow Functions: filter()
    k. ES6 - Arrow Functions: find() and findIndex()
    l. ES6 - Introduction to Classes
    m. ES6 - Inheritance in ES6
    n. ES6 - Callbacks and Promises
    o. ES6 - Working with Promises
    p. ES6 - Chaining Promises
2. Introduction to React JS
    a. What is React?
    b. Why React?
    c. React version history
    d. Just React - Hello World
    e. Using create-react-app
    f. Anatomy of react project
    g. Running the app
    h. Debugging first react app
    i. Install create-react-app and create a new react project.
3. Templating with JSX
    a. Working with Rect.create element
    b. Expressions

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   c. Using logical Operators
   d. Specifying attributes
   e. Specifying children"
4. React Components
   a. Significance of component architecture
   b. Types of components
   c. Functional
   d. Class based
   e. Pure
   f. Component Composition
   g. Create class based and functional components.
5. State and Props
   a. What is state and its significance
   b. Read state and set state
   c. Passing data to component using props
   d. Validating props using PropTypes
   e. Supplying default values to props using defaultProps
   f. Create a stateful component and stateless component. Pass data from parent component to child component using props. Implement child to parent communication using callbacks.
6. Lists
   a. Using react key prop
   b. Using map function to iterate on arrays to generate elements
   c. Create component which renders lists iteratively using map function of array.
7. More about components
   a. Component life cycle
   b. Events
   c. Managing errors
   d. Work with events to set state properties
   e. Learn to use Error Boundary component to identify errors in child components using map function of array
8. Forms
   a. Controlled Form Components
   b. Uncontrolled Form Components
   c. Handling inputs efficiently
   d. Learn to use an input element as a controlled component
9. Props
   a. Render Props
   b. Higher Order Components
   c. Build a Currency-Converter component that implements render props.
10. Portals

   a. Portal's introduction
   b. Event bubbling
   c. Use Portals to render a dialog box component in a DOM element.
11. Global and shared data
   a. Unidirectional Data Flow
   b. Challenges with Props
   c. Context APIs
   d. Instantiate a context, and provider and consumer components to trickle data down to the component hierarchy
12. Hooks
   a. Introduction to Hooks
   b. The useState hook
   c. The useContext hook
   d. The useReducer hook
   e. Use the useEffect function to fetch data from a stock app
   f. Implement the useContext hook
   g. Use the useReducer hook
   h. Build a custom hook
13. Routing
   a. Setting up react router
   b. Understand routing in single page applications
   c. Working with BrowserRouter and HashRouter components
      i. Configuring route with Route component
   d. Using Switch component to define routing rules
   e. Making routes dynamic using route params
   f. Working with nested routes
   g. Navigating to pages using Link and NavLink component
   h. Redirect routes using Redirect Component
   i. Using Prompt component to get consent of user for navigation
   j. Path less Route to handle failed matches
   k. Understand nested routing
   l. Learn to protect a route
14. Code Splitting
   a. Code splitting and Suspense
   b. Learn to us React.lazy and Suspense to load two components
15. Isomorphic React
   a. Server-Side Rendering
   b. SSR with React
16. Redux
   a. What is redux
   b. Why redux

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   c. Redux principles
   d. Install and setup redux
   e. Creating actions, reducer and store
17. Immutable.js for immutable data structures
   a. What is Immutable.js?
   b. Immutable collections
   c. Lists
   d. Maps
   e. Sets
18. React Redux
   a. What is React Redux
   b. Why React Redux
   c. Install and setup
   d. Presentational vs Container components
   e. Understand high order component
   f. Understanding mapStateToProps and mapDispatchtToProps usage"
19. Redux Saga
   a. Why redux middleware
   b. Available redux middleware choices
   c. What is redux saga
   d. Install and setup redux saga
   e. Working with Saga helpers
   f. Sagas vs promises
20. Testing
   a. Introduction to Testing
   b. Using Jest with Test Utils from React-DOM
   c. Using Jest with the React Testing Library
   d. Using Jest with Enzyme

Project 1: Building an event finder app

TYPESCRIPT

1. Introduction
   a. What is Typescript?
   b. Benefits of Typescript
   c. Setup the Environment
   d. First Typescript Example"
2. Data Types and Variables
   a. Basic Data Types
   b. Arrays
   c. Tuples
   d. Enum

   e. Any and void
   f. Null and undefined
   g. Type Inference
   h. Type Casting
   i. Difference between let and var
   j. Const declaration
3. Destructuring & Spread
   a. Array Destructuring
   b. Object Destructuring
   c. Mixed Destructuring
   d. Property renaming
   e. Default Values
   f. Spreads
4. Working with Classes
   a. Writing and Using Classes
   b. Constructor method
   c. Inheritance of classes
   d. Type casting
   e. Type Assertion
   f. Static Properties
   g. Abstract class
5. Working with Interfaces
   a. Interface Declaration and Initialization with an object
   b. Duck Typing
   c. Interface Implementation by class
   d. Interface having Optional Property
   e. Class extending another class and also implementing an Interface
   f. Excess Property Checks
   g. Indexable Types
   h. Extending Interfaces
6. Function Types
   a. Optional and Default Parameters
   b. Rest Parameters
   c. Function Overloading
   d. Function Types"
7. Modules
   a. Introduction
   b. Exporting and Importing of Modules
   c. Re-Export of Modules
   d. Default Exports
8. Namespaces

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   a. Introduction
   b. Syntax with Example
   c. Compiling multiple TS files as one JS file"
9. Ambients
   a. Definition
   b. Syntax
   c. Examples

Project 2: Demo site

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